

MODULE 1: INTRODUCTION TO OPERATIONS RESEARCH

Theoretical Questions

Year & Q.No	Marks	Question Text	Type
2024 Q.1(a)	3 Marks	Define Operations Research. REPEATED	Theory
2023 Q.1(a)	3 Marks	What is Operations Research? REPEATED	Theory
2023 Q.2(b)	7 Marks	Enumerate on the scope of Operations Research.	Theory
2025 Q.1(a)	3 Marks	Give a brief historical development of Operations Research.	Theory
2025 Q.2(b)	7 Marks	Define the term Operations Research. Mention different phases of Operations Research. REPEATED	Theory

MODULE 2: LINEAR PROGRAMMING PROBLEM (LPP) & GRAPHICAL METHOD

Theory Questions

Year & Q.No	Marks	Question Text	Type
2024 Q.1(b)	7 Marks	Give applications of Linear Programming Problem in management. REPEATED	Theory
2024 Q.2(a)	3 Marks	What are the limitations in LPP?	Theory
2024 Q.5(a)	3 Marks	What are the limitations of graphical approach to solve LPP?	Theory
2023 Q.2(a)	3 Marks	What are the applications of LPP in management? REPEATED	Theory

Practical / Numerical Problems

Year & Q.No	Marks	Problem Description	Type
2024 Q.1(c)	10 Marks	LPP Formulation: Mutual fund with Rs. 20 lakhs investing in Govt bonds, blue chip stocks, speculative stocks, short-term deposits. Maximize expected annual return.	Practical
2024 Q.2(c)	10 Marks	Graphical Method: Minimize $Z = 20x_1 + 10x_2$ subject to $x_1 + 2x_2 \leq 40$, $3x_1 + x_2 \geq 30$, $4x_1 + 3x_2 \geq 60$ where $x_1, x_2 \geq 0$. VERBATIM REPEAT	Practical
2023 Q.1(b)	7 Marks	LPP Formulation: Manufacturer producing models M_1 and M_2 with grinding and polishing machine hour constraints. Maximize profit.	Practical
2023 Q.3(c)	10 Marks	Graphical Method: Minimize $Z = 20x + 10y$ subject to $x + 2y \leq 40$, $3x + y \geq 30$, $4x + 3y \geq 60$ where $x, y \geq 0$. VERBATIM REPEAT	Practical
2025 Q.1(b)	7 Marks	LPP Formulation: Firm producing products A & B using raw material (100 kg) and processing time (200 hrs). Formulate LPP to maximize profit.	Practical
2025 Q.1(c)	10 Marks	Graphical Method: Minimize $Z = 6x_1 + 8x_2$ subject to $x_1 + x_2 \leq 450$, $2x_1 + x_2 \leq 600$ and $x_1, x_2 \geq 0$.	Practical

MODULE 3: DECISION THEORY & JOB SEQUENCING

Theory Questions

Year & Q.No	Marks	Question Text	Type
2024 Q.2(b)	7 Marks	Write down different steps in Decision Making process. REPEATED	Theory
2024 Q.7(a)	3 Marks	Write any three assumptions under sequencing problem. REPEATED	Theory
2023 Q.5(b)	7 Marks	Enumerate on the different criteria of decision making under uncertainty.	Theory
2023 Q.6(a)	3 Marks	Why is job sequencing important? REPEATED	Theory

Year & Q.No	Marks	Question Text	Type
2023 Q.6(b)	7 Marks	Briefly explain the steps involved in decision-making process. REPEATED	Theory
2025 Q.2(a)	3 Marks	State the basic assumptions underlying sequencing problems. REPEATED	Theory
2025 Q.5(a)	3 Marks	Write a short note on Decision Theory.	Theory
2025 Q.6(a)	3 Marks	Define the term sequencing and state its importance. REPEATED	Theory

Practical / Numerical Problems

Year & Q.No	Marks	Problem Description	Type
2024 Q.4(c)	10 Marks	n Jobs on 3 Machines: 5 Jobs (J_1 to J_5) through machines M_1 , M_2 , M_3 . Find optimal sequence, total elapsed time, and idle times.	Practical
2023 Q.1(c)	10 Marks	n Jobs on 3 Machines: 4 Jobs on Shaping, Drilling, and Tapping machines. Determine optimal sequence, total elapsed time, and total idle time.	Practical
2025 Q.2(c)	10 Marks	n Jobs on 2 Machines: 5 Jobs (J_1 to J_5) through machines M_1 and M_2 . Find optimal job sequence, total elapsed time, and idle time.	Practical
2025 Q.6(c)	10 Marks	n Jobs on 3 Machines: 4 Jobs (J_1 to J_4) through machines M_1 , M_2 , M_3 . Obtain optimal sequence and processing times.	Practical

MODULE 4: TRANSPORTATION PROBLEMS

Theory Questions

Year & Q.No	Marks	Question Text	Type
2024 Q.3(a)	3 Marks	What is an unbalanced case in a transportation model? REPEATED	Theory
2023 Q.7(a)	3 Marks	What is Degeneracy in transportation problem? REPEATED	Theory
2025 Q.3(a)	3 Marks	What is meant by an unbalanced transportation problem? REPEATED	Theory
2025 Q.4(a)	3 Marks	Write a note on degeneracy of transportation problem. REPEATED	Theory

Practical / Numerical Problems & Case Studies

Year & Q.No	Marks	Problem Description	Type
2024 Q.3(b)	7 Marks	North-West Corner Method (NWCM): Illustrate basic feasible solution for 4x4 transportation matrix.	Practical
2024 Q.3(c)	10 Marks	Vogel's Approximation Method (VAM): Find initial basic feasible solution for 3x4 transportation matrix.	Practical
2024 Q.7(c)	10 Marks	Optimality Testing: Given a solved transportation solution table, test if the solution is optimal. Justify your answer.	Practical
2023 Q.2(c)	10 Marks	Maximization Transportation Problem: Solve to maximize profit by Least Cost Method for 3x4 matrix.	Practical
2023 Q.8 (Case)	20 Marks	Compulsory Case Study: 3x4 Factory-to-Warehouse matrix. (i) Solve by VAM and check optimality by MODI method, (ii) Check if optimal, (iii) Check for alternate optimal solutions.	Case Study
2025 Q.3(b)	7 Marks	North West Corner Rule (NWCR): Find initial basic feasible solution to transportation problem.	Practical
2025 Q.5(b)	7 Marks	Vogel's Approximation Method (VAM): Obtain initial basic feasible solution for transportation matrix.	Practical
2025 Q.8 (Case)	20 Marks	Compulsory Case Study: Larsen & Toubro demand at 4 sites vs supply at 3 places. Determine optimum transportation schedule and optimal cost.	Case Study

MODULE 5: THEORY OF GAMES & ASSIGNMENT PROBLEMS

Theory Questions

Year & Q.No	Marks	Question Text	Type
2024 Q.4(a)	3 Marks	What is Saddle point in game theory?	Theory
2023 Q.3(a)	3 Marks	What is an Assignment problem?	Theory
2025 Q.5(c)	10 Marks	Define terms in Game Theory: (i) Payoff matrix, (ii) Pure strategy, (iii) Mixed strategy, (iv) Value of Game, (v) Fair Game.	Theory
2025 Q.6(b)	7 Marks	Explain the Max. Min and Min. Max principles with an example for each.	Theory

Practical / Numerical Problems

Year & Q.No	Marks	Problem Description	Type
2024 Q.4(b)	7 Marks	Game Theory Strategy: Company management & labor union negotiations (4x4 payoff matrix). Determine optimal strategies and value of game.	Practical
2024 Q.5(b)	7 Marks	Game Theory Graphical Method: Solve 2x4 game matrix for Player A & B strategies.	Practical
2024 Q.6(b)	7 Marks	Assignment Problem: Assign 3 programmers to 3 programs to minimize total computer time.	Practical
2024 Q.6(c)	10 Marks	Game Theory Algebraic Method: Solve 2x3 game problem for Player A & B.	Practical
2024 Q.8(b)	10 Marks	Restricted Assignment Problem: Assign machines $M_1..M_4$ to positions A..E where M_2 cannot be placed at C and M_3 cannot be placed at A.	Case Part
2023 Q.3(b)	7 Marks	Principle of Dominance: Solve 4x4 payoff matrix for Company X & Company Y.	Practical
2023 Q.4(c)	10 Marks	Travelling Salesman Problem: Solve travelling salesman problem for given distance/ cost matrix between 5 cities.	Practical
2023 Q.6(c)	10 Marks	Game Theory Graphical Method: Solve 2x5 game matrix graphically.	Practical
2023 Q.7(b)	7 Marks	Assignment Problem: Solve 4x4 assignment cost matrix to minimize cost.	Practical
2025 Q.3(c)	10 Marks	Assignment Problem: Assign tasks (1, 2, 3) to persons (A, B, C) to minimize total cost.	Practical
2025 Q.4(b)	7 Marks	Principle of Dominance: Solve 3x3 game matrix using concept of dominance.	Practical
2025 Q.4(c)	10 Marks	Game Theory Graphical Method: Solve 2x3 game using graphical method.	Practical

MODULE 6: PROJECT MANAGEMENT (PERT & CPM NETWORKS)

Theory Questions

Year & Q.No	Marks	Question Text	Type
2024 Q.6(a)	3 Marks	What are Looping and Dangling errors in Networking?	Theory
2024 Q.7(b)	7 Marks	Distinguish between PERT and CPM method. REPEATED ALL 3 YEARS	Theory
2023 Q.4(a)	3 Marks	What do you understand by "work-breakdown" structure?	Theory
2023 Q.4(b)	7 Marks	Enumerate on the differences between PERT and CPM. REPEATED ALL 3 YEARS	Theory
2023 Q.5(a)	3 Marks	What is "Project control phase" in project management?	Theory
2025 Q.7(a)	3 Marks	Describe the phases of project management.	Theory
2025 Q.7(b)	7 Marks	Differentiate between PERT and CPM. REPEATED ALL 3 YEARS	Theory

Practical / Numerical Problems

Year & Q.No	Marks	Problem Description	Type
2024 Q.5(c)	10 Marks	CPM Network Analysis: Construct network for activities A to I with durations. Find critical path, critical activities, and total project completion time.	Practical
2024 Q.8(a)	10 Marks	CPM Network Floats: From activity predecessor table, draw network diagram and calculate ES, EF, LS, LF, and Total Float.	Case Part
2023 Q.5(c)	10 Marks	CPM Network & Floats: Draw network diagram for activities A to F. Find (i) Critical path, (ii) Total floats available on non-critical activities.	Practical
2023 Q.7(c)	10 Marks	PERT 3-Time Estimates: Given optimistic, most likely, pessimistic times for activities 1-2 to 7-8: (i) Draw network, (ii) Calculate expected times, (iii) Determine critical path, (iv) Probability to finish in 4 years, (v) Probability to finish in 55 months.	Practical
2025 Q.7(c)	10 Marks	CPM Network Analysis: Small project with jobs 1-2 to 6-7 and durations. (i) Draw arrow diagram, (ii) Find critical path and total project duration.	Practical